

Expanding Brackets Using a Grid

Example

Expand and simplify: $(x + 4)(x + 2)$

Step 1

Draw out a 3 x 3 multiplication grid.

Step 2

Fill in your grid with the terms from the brackets (don't forget to include the signs).

	x	$+ 4$
x		
$+ 2$		

Step 3

Now multiply the terms.

	x	$+ 4$
x	x^2	$+ 4x$
$+ 2$	$+ 2x$	$+ 8$

Step 4

You have now fully expanded your double brackets. Your next step is to write out the terms you have calculated.

$$x^2 + 2x + 4x + 8$$

Step 5

Simplify your answer:

Your final answer is $x^2 + 6x + 8$

